

Measuring the Charge-diffusion Kernel of a Back-illuminated CCD in Orbit from Glancing Particle Tracks

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Abstract.

The lateral diffusion of photogenerated charge in the undepleted layer of a back-illuminated silicon sensor sets its point spread function and the statistics of its noise in the ultraviolet, but it is rarely measured on a flight sensor. We show that glancing particle tracks, which cross the full thickness of the sensor over a length of tens of pixels, sample the diffusion width at every depth in a single exposure, and we use them to measure the depth-dependent charge-diffusion kernel of the three CCDs of the Interface Region Imaging Spectrograph in orbit, from 1070 tracks found in level-1 images taken mostly inside the South Atlantic Anomaly. The width of the charge cloud at the back surface, the depth at which it vanishes, and the probability that two electrons deposited at the same depth are collected in the same pixel are measured directly and compared with a field-free diffusion model. On the Slit-Jaw Imager the same-pixel probability at the back surface is 0.34 ± 0.01 , against 0.34 from the model, and the tracks reveal a residual spread of about a micron inside the depletion region which the model neglects. The method needs no laboratory access and applies to any back-illuminated sensor in orbit.

Keywords: charge-coupled devices, charge diffusion, point spread function, ultraviolet detectors, particle tracks, IRIS.

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1 Introduction

A back-illuminated silicon charge-coupled device (CCD) absorbs ultraviolet (UV) light within a fraction of a micron of its back surface, in a layer where there is no electric field to sweep the photogenerated charge toward the gates. The charge diffuses until it reaches the edge of the depletion region, and the resulting charge cloud, which can be several microns wide, is shared among neighboring pixels. This sharing broadens the point spread function (PSF) of the sensor¹ and, because the electrons liberated by one photon are shared together, it changes the statistics of the image noise, entering the variance-to-mean ratio (VMR) of a photon-transfer curve through the probability that two electrons from the same photon land in the same pixel.²

The width of the charge cloud is set by the thickness of the undepleted layer, which depends on the resistivity of the wafer and the bias applied to the device, both of which vary between nominally identical sensors and are rarely known for a sensor in orbit. Laboratory measurements use light of varying penetration depth^{3,4} or the split events of soft X-rays,⁵ and neither is available once the instrument has launched.

Energetic particles offer a probe that arrives for free. A particle crossing the sensor at glancing incidence deposits charge along a track tens of pixels long, and because it enters at the back surface and exits at the gates, the fractional position along the track is the fractional depth of the charge. Cosmic-ray muons have been used in this way to measure the growth of charge diffusion with drift distance in thick, fully depleted CCDs,⁶ where the cloud widens with depth. In a thin sensor with an undepleted back surface the dependence is reversed, the cloud is widest at the back surface and collapses to a pixel-sharp line in the depletion region, and the same tracks measure the field-free kernel that governs UV imaging.

Here we apply this method to the three CCDs of the Interface Region Imaging Spectrograph (IRIS),⁷ whose far ultraviolet (FUV) and near-UV photon-transfer curves⁸ motivated the question. Section 2 describes the tracks, the model of their cross-section, and the fit; Section 3 presents the kernel of each sensor; and Section 4 compares it with the field-free model and discusses the systematics, including the charge density of the tracks.

2 Method

2.1 Data

IRIS⁷ carries three CCDs of the same design, two behind the spectrograph (FUV1 and FUV2, which share one level-1 image) and one behind the Slit-Jaw Imager (SJI). They are back-illuminated, 14 μm thick, with 13 μm pixels, and their level-1 images are the raw frames with only the dark and the readout corrections applied, so a particle hit appears exactly as the camera recorded it. We searched 3249 such images, 680 of them taken inside the South Atlantic Anomaly (SAA), where the flux of trapped protons at the altitude of IRIS is greatest,^{9,10} drawn from the five campaigns of Table 1. Figure 1 shows a spectrograph image and a slit-jaw image taken seconds apart during one pass through SAA: the particle hits are dense, and a handful of them are the long glancing tracks we want. Tracks were sought against the dark background of the spectrograph frames and, for SJI, in the off-limb half of the field of view.

Table 1 The observing campaigns searched for tracks. The particle population is inferred from the energy loss of the tracks and from whether the frames were taken inside SAA; the number of frames inside SAA is given in parentheses. The core tracks are the flat tracks with $0.25 < t_c < 0.6$, over which the mean t_c with its standard error and the median σ_{max} with its interquartile range are taken.

date	particles	roll	frames	tracks	flat	core	t_c	σ_{max} (μm)
2014-03-12, FUV	SAA protons	0°	443 (57)	220	76	53	0.430 ± 0.008	5.5 [5.5, 6.0]
2014-04-07, FUV	cosmic rays	0°	975 (0)	49	21	16	0.416 ± 0.016	5.5 [4.5, 5.5]
2018-03-16, FUV	cosmic rays	0°	311 (17)	85	40	29	0.395 ± 0.012	5.0 [4.5, 5.5]
2018-05-04, FUV	SAA protons	−90°	362 (137)	395	140	97	0.429 ± 0.008	5.5 [4.5, 6.5]
2018-05-04/06, SJI	SAA protons	−90°	1158 (469)	321	140	113	0.430 ± 0.006	5.5 [5.0, 6.0]

2.2 Finding tracks

The background of each frame was estimated as the trimmed mean of the frames in the same observing sequence and subtracted, and the read noise was estimated from the median absolute deviation of the residual. Pixels more than five times the read noise above the background were labeled, and a straight line was fit by least squares to the charge-weighted centroid of each row of every connected group of labeled pixels. We kept the groups spanning at least twelve rows (or columns) with a slope below 0.35 pixels per row, so that each track is nearly aligned with the pixel grid and its cross-section is sampled once per row without resampling the pixels, and extracted a seven-pixel-wide cutout centered on the integer part of the fitted line in every row. This yielded 1070 tracks. The slope cut is not as costly as it sounds: Figure 2 shows that the trapped protons arrive strongly aligned with the slit axis, so most of them pass it.

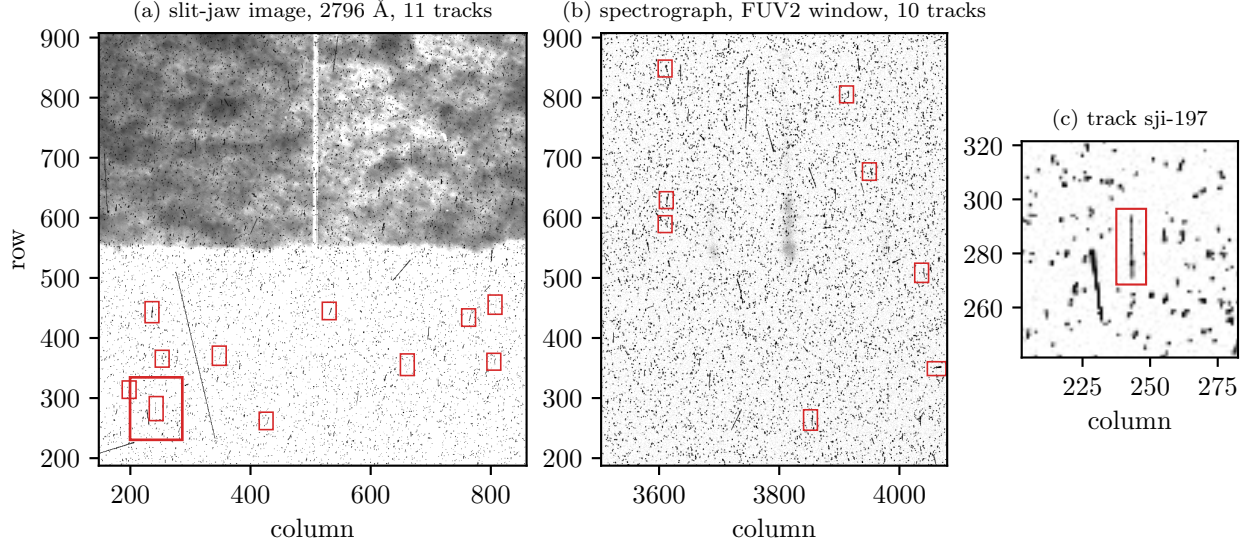


Fig 1 Level-1 images taken seconds apart during a pass through SAA on 2018 May 4, with the spacecraft rolled by -90° so that the limb runs horizontally across the field. (a) The slit-jaw image at 2796 Å, whose lower half is off the limb. (b) The widest of the readout windows of the spectrograph image, on the FUV2 CCD. (c) One of the tracks in (a) at full resolution. The images are shown with an inverse hyperbolic sine stretch above the background, saturating at 400 data numbers, and the red boxes outline the tracks the finder kept from each frame. Most of the particle hits are the short streaks and dots that pepper both images; the tracks used here are the long, thin, nearly vertical ones.

2.3 Model

Figure 3 shows the principle. A particle that enters the back surface and exits the front surface spans the full thickness of the sensor, so the fractional depth of slice i of an N -slice track is $t = (i + 1/2)/N$, up to the orientation of the track, and the charge deposited in each slice diffuses from a known depth. We model the charge in every slice as spread uniformly along the fitted centerline and diffused as a Gaussian with standard deviation

$$\sigma(t) = \sigma_{\max} \sqrt{1 - t/t_c}, \quad t < t_c, \quad (1)$$

and zero otherwise, where $t_c = z_f/D$ is the fractional thickness of the field-free region and $\sigma_{\max}$ is the width of the charge cloud at the back surface. This is the depth dependence of a random walk to an absorbing plane at z_f , the form used to model the UV noise of these sensors in the companion article, evaluated here in fractional depth because we observe the length of a track in slices rather than the thickness of the sensor in microns. The uniform spread along the tilted centerline within a slice is approximated by adding $m^2/12$ to the variance, where m is the slope in pixels per row.

We fit the orientation, t_c , and $\sigma_{\max}$ by exhaustive search on a grid of 21 values in each of $t_c \in [0, 1]$ and $\sigma_{\max} \in [0, 10] \mu\text{m}$. At each grid point the offset and tilt of the centerline are adjusted, by up to 0.6 pixels and 0.03 pixels per row, to minimize a robust misfit, $\sum \ln(1 + r^2/2)$, where r is the residual of the fraction of each slice's charge in each pixel in units of the read noise. This is the negative log-likelihood of a Cauchy distribution rather than a Gaussian, so that a single stray pixel, from a delta ray or a second hit inside the cutout, cannot dominate the fit.

2.4 Selection

The useful tracks are those which cross the full thickness of the sensor at nearly constant energy loss. We kept only the tracks for which the fit constrains t_c to within 0.15 and improves on a model with no diffusion by at least ten units of misfit, and for which the median charge per slice in the last third of the track is within 50% of that in the first third, since a rise in the deposited charge along the track is the Bragg peak of a particle that stopped inside the sensor and did not cross its full thickness. We call these the flat tracks; 417 tracks pass these cuts, 140 of them on the SJI CCD, and Figure 8 shows the longest of them alongside a stopping track.

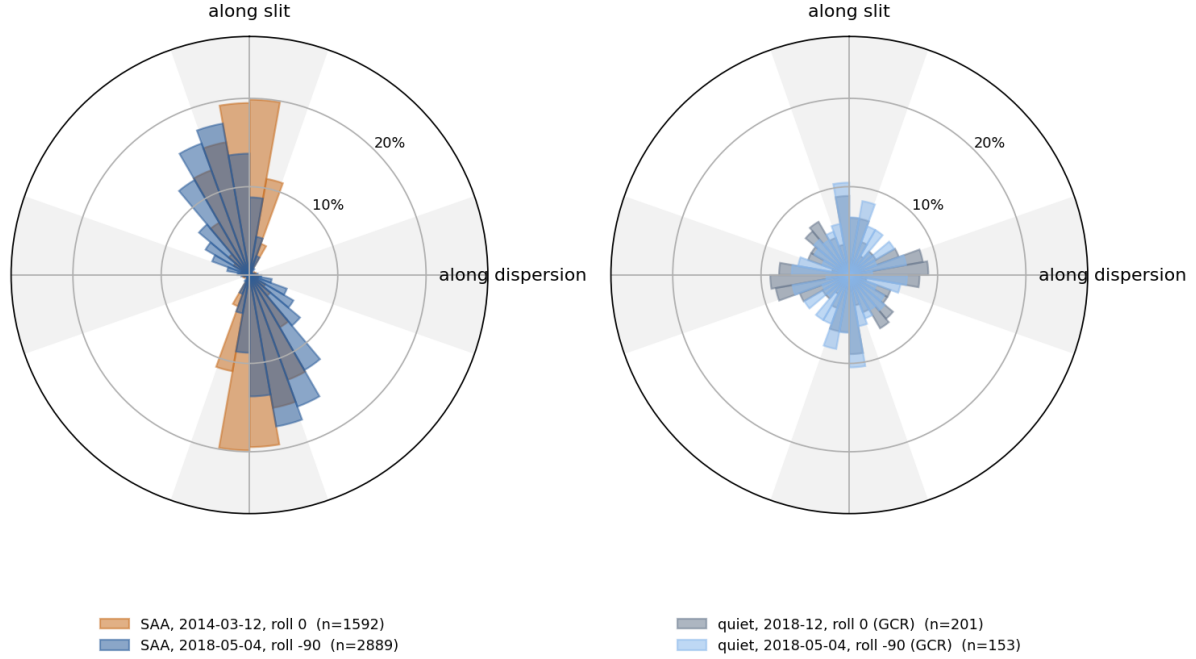


Fig 2 The azimuth in the detector frame of every elongated connected component above 5σ in the spectrograph frames, with a charge-weighted width under 3 pixels and an aspect ratio over 4, before any of the finder's cuts, so that the whole circle is populated. Tracks are undirected, so each is drawn at θ and $\theta + 180^\circ$, and the grey wedges are the windows about the row and column axes that the finder accepts. Left: the frames taken inside SAA at both spacecraft rolls. Right: quiet frames at the same rolls. The trapped protons arrive strongly aligned, mostly within the window along the slit, while the cosmic rays are isotropic to within counting noise; the preferred direction shifts and broadens when the spacecraft is rolled, so it is a property of the arrival direction as seen by the instrument rather than of the detector.

3 Results

3.1 The width against depth

Figure 4 shows the flat tracks stacked on their fitted centerlines and the diffusion width they imply at each depth, measured two ways. The first assumes no depth dependence at all: for each depth bin, the misfits of every slice in the bin are summed over a grid of trial widths and the width which minimizes the sum is taken, so the wedge shape emerges from the data rather than being imposed. The second is the average of the per-track fits of Equation 1. The two agree to within a few tenths

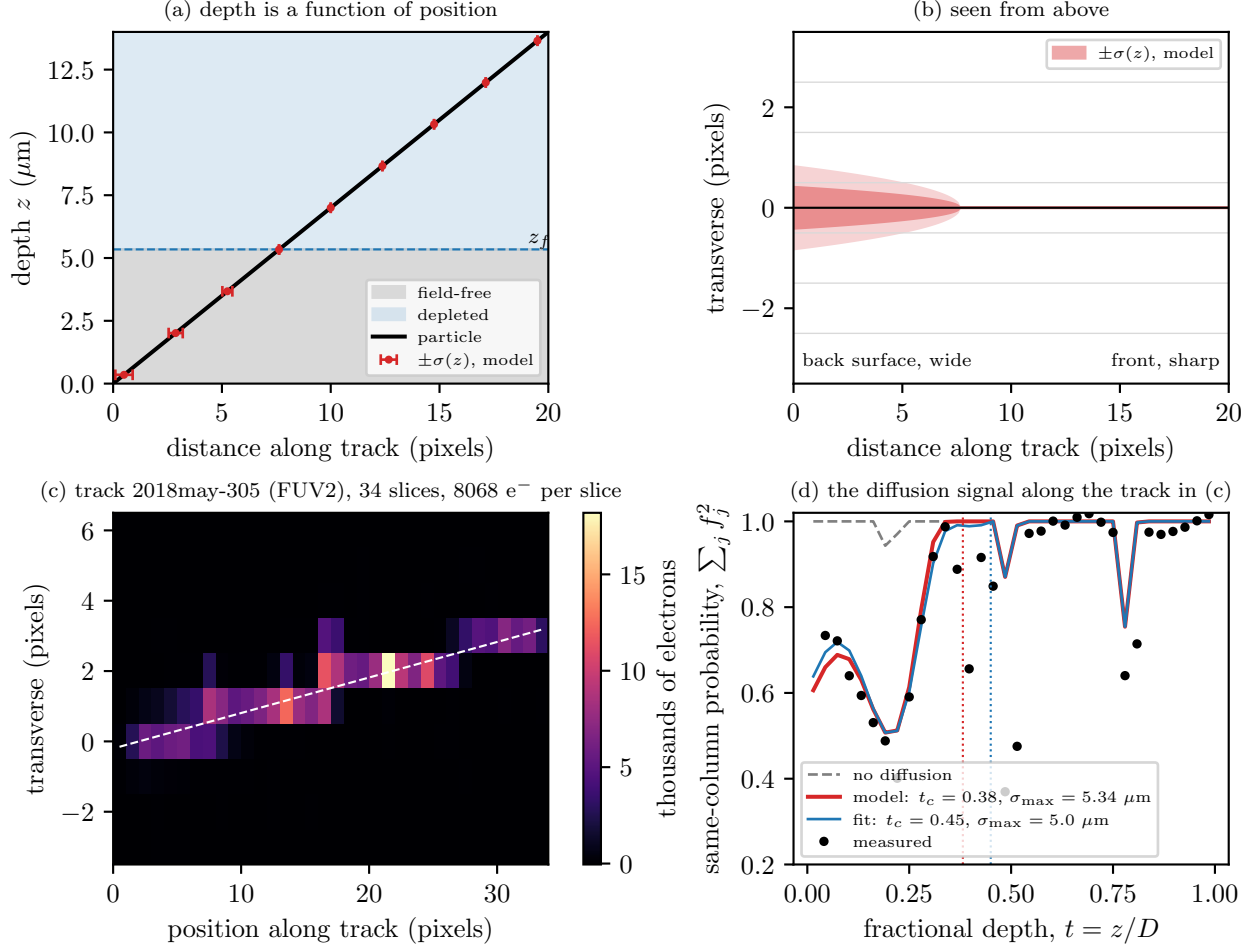


Fig 3 How a glancing track measures the diffusion width against depth. (a) A particle entering at the illuminated back surface ($z = 0$) and leaving at the gates ($z = D$) crosses the field-free layer first, where the charge it liberates diffuses until it reaches the depletion edge, so the lateral spread (red bars, to scale in pixels) is largest at the back surface and vanishes at $z = z_f$. (b) Seen from above, the track is a wedge: about 0.4 pixels wide at one end and pixel-sharp beyond t_c . (c) A proton track on the FUV2 CCD, in the coordinates of its parent image, with the fitted centerline dashed. (d) The probability that two electrons deposited in the same slice of that track are collected in the same column, $\sum_j f_j^2$ corrected for read noise, for every slice, against the field-free model, the best fit, and a model with no diffusion, all evaluated at the fitted centerline. The dip near $t = 0.2$ is where the centerline crosses a pixel boundary; the diffusion signal is the gap between the measured points and the no-diffusion curve.

of a micron on all three CCDs except beyond t_c , where the model-free profile does not go to zero but keeps a floor of 0.5 to 1 μm across the depletion region, which the two-parameter model cannot represent.

3.2 The fitted parameters

Figure 5 shows the fitted t_c and $\sigma_{\max}$ of every flat track, and Table 1 their means in each campaign. The critical depth peaks at 0.40 to 0.50 on all three CCDs, to the right of the 0.38 of the field-free model, and the two particle populations, the two spacecraft rolls, and the two cameras agree to within a few hundredths. The back-surface width does differ between chips: FUV2 and SJI cluster

at 5.5 to 6.5 μm while FUV1 clusters at 3 to 5 μm , so the pooled scatter is mostly this chip-to-chip difference, which is expected from the resistivity of the wafer and the applied bias and is one reason a sensor's kernel is worth measuring rather than assuming.

3.3 The same-pixel probability

For the noise model the quantity that matters is not the width but the probability that two electrons deposited at the same depth are collected in the same pixel. For a slice with pixel fractions f_j that is $p = \sum_j f_j^2$, corrected for the read-noise contribution $\sum_j \epsilon_j^2$, and it can be read directly off each slice with no model of the shape of the kernel. Since the kernel is separable, the probability that two electrons are collected in the same pixel is $\mathcal{P} = p^2$. Figure 6 shows p against depth on each CCD, and Table 2 lists it for the slices within $D/10$ of the back surface, alongside the prediction of the field-free model evaluated at the fitted centerline of every track. On the SJI CCD the back surface gives $\mathcal{P} = 0.34 \pm 0.01$ against 0.34 from the model, and FUV2 agrees as well, while FUV1 spreads its charge over fewer pixels. Beyond t_c the measured probability settles at about 0.85 rather than the 0.95 of a perfectly sharp cloud on all three CCDs, the same floor seen in the widths.

3.4 Diffusion inside the depletion region

To quantify that floor we add a third parameter to the model,

$$\sigma^2(t) = \sigma_{\max}^2 \left(1 - \frac{t}{t_c}\right) + \sigma_d^2 g(t), \quad (2)$$

where the first term applies only for $t < t_c$, and $g = 1$ in the field-free layer, since that charge drifts across the full depleted thickness, falling linearly to zero at the gates. The grid is t_c from 0.25 to 0.60, $\sigma_{\max}$ from 2 to 8 μm , and σ_d from 0 to 3 μm , with the same centerline nuisance parameters as before, applied to every flat track with its orientation taken from the two-parameter fit. Figure 7 shows the result. The misfit pooled over the tracks on each CCD has a sharp minimum at $\sigma_d = 1.0$ μm , with 1.5 μm already strongly disfavored; the improvement is hundreds of units of misfit pooled but only one or two per track, and the value each track prefers on its own is spread over 0 to 1 μm , so the effect is modest for any one track and consistent across them. Adding it shifts the other two parameters: t_c drops toward the model value and $\sigma_{\max}$ rises by about half a micron, since some of what the two-parameter fit attributed to the field-free layer was really this floor. The resolution of the centerline matters here: a centerline located only to 0.1 pixel is uncertain by 1.3 μm , the size of the effect being sought, and with such a grid the minimum washes out entirely.

Table 2 The charge diffusion measured from glancing particle tracks on the three IRIS CCDs. The number of frames is the number of level-1 images searched, with the number taken inside SAA in parentheses. The flat tracks are those which constrain t_c to within 0.15 and show no Bragg rise along their length. t_c and $\sigma_{\max}$ are the medians of the per-track fits, with the interquartile range in brackets. p is the probability that two electrons deposited within $D/10$ of the back surface are collected in the same column, $\mathcal{P} = p^2$ is the corresponding probability that they are collected in the same pixel, and the last column is $\mathcal{P}$ predicted by the field-free model for the same tracks.

CCD	frames	tracks	flat	t_c	$\sigma_{\max}$ (μm)	p	$\mathcal{P}$	$\mathcal{P}_{\text{model}}$
FUV1	2091 (211)	298	106	0.45 [0.35, 0.65]	4.5 [3.0, 5.5]	0.647 ± 0.025	0.419 ± 0.033	0.325
FUV2	2091 (211)	451	171	0.45 [0.40, 0.55]	5.5 [5.0, 6.5]	0.577 ± 0.012	0.332 ± 0.013	0.331
SJI	1158 (469)	321	140	0.45 [0.40, 0.50]	5.5 [5.0, 6.0]	0.581 ± 0.009	0.338 ± 0.010	0.340

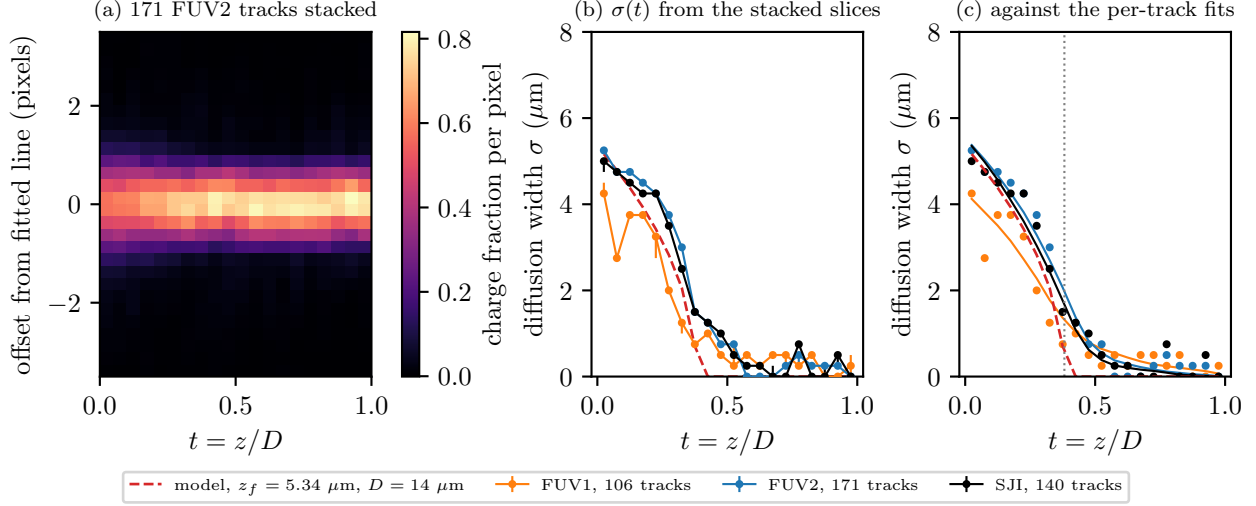


Fig 4 (a) Every slice of every flat track on the FUV2 CCD placed at its fractional depth and aligned on its fitted centerline, with each pixel's charge spread uniformly over its own width. The wedge is subtle: the wings at $t < 0.4$ carry a few percent of the charge. (b) The diffusion width in each depth bin, found by summing the misfit of every slice in the bin over a grid of trial widths and minimizing, with no parametric model of the depth dependence. Error bars span the widths within two units of misfit of the minimum. (c) The same points against the average of the per-track fits of Equation 1, which reproduces the model-free profile except beyond t_c , where the tracks keep a floor of 0.5 to $1 \mu\text{m}$ that the model sets to zero.

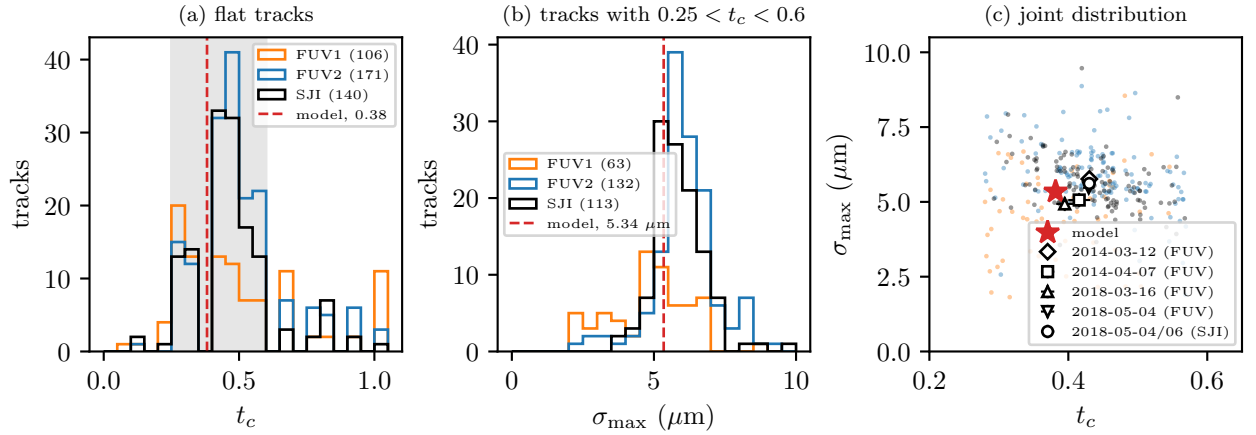


Fig 5 The fitted parameters. (a) The critical depth of every flat track on each CCD, with the field-free model dashed. The pile-up at $t_c = 1$ and the tail below 0.25 are tracks the two-parameter model cannot describe, diffuse or sharp along their whole length, and they are excluded from the shaded core used elsewhere. (b) The back-surface width of the core tracks. (c) Each core track jittered off the fit grid, colored by CCD as in (a), with the mean of each dataset and its standard error as open symbols. Two particle populations, two spacecraft rolls, and two cameras agree to within a few hundredths in t_c , all above the model, and the diagonal smear is the degeneracy between t_c and σ_{max} in a single-track fit.

4 Discussion

At the back surface, where UV photons are absorbed, the tracks confirm the field-free model on the SJI and FUV2 CCDs: the same-pixel probability of 0.34 ± 0.01 on SJI matches the 0.34 predicted

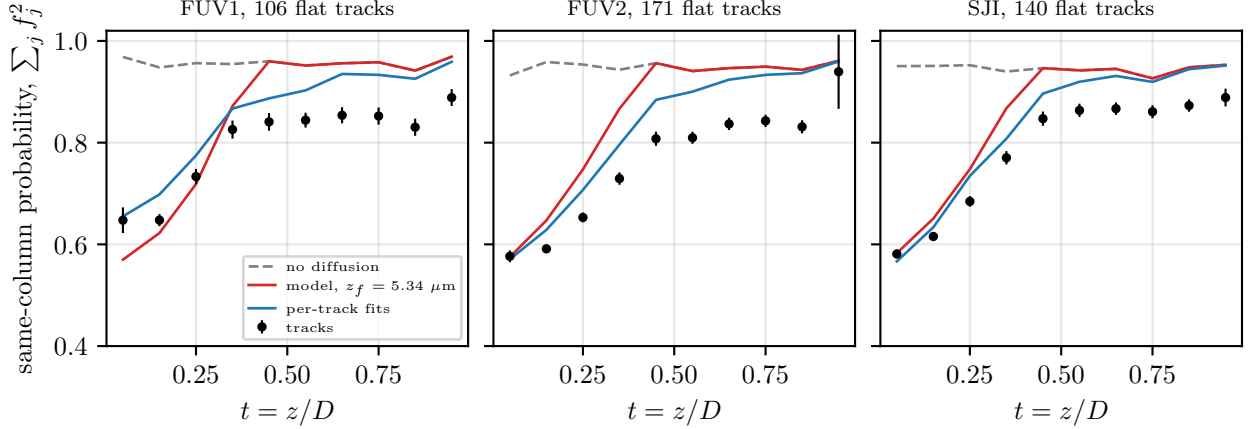


Fig 6 The probability that two electrons deposited in the same slice are collected in the same column, averaged over the flat tracks on each CCD in bins of fractional depth, with the standard error of each mean. The red line is the field-free model, Equation 1 with $t_c = 0.38$ and $\sigma_{\max} = 5.34 \mu\text{m}$, evaluated at the fitted centerline of each track and averaged in the same bins; the blue line is the same for each track's own fit; and the dashed line is the same with no charge diffusion, which is below one only where a centerline runs near a pixel boundary. The back surface agrees with the model on the FUV2 and SJI CCDs, while beyond t_c the tracks settle below the no-diffusion curve on all three.

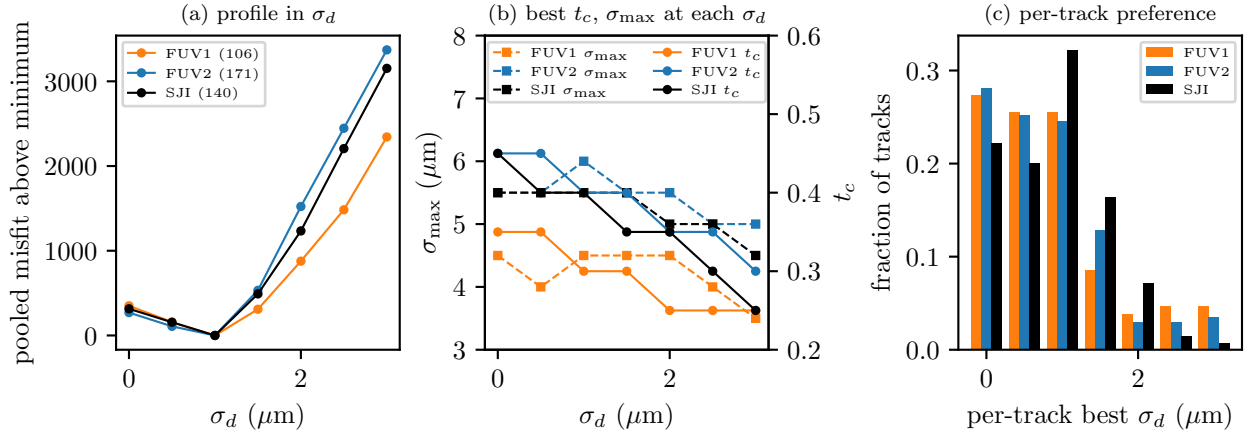


Fig 7 Allowing diffusion inside the depletion region, Equation 2. (a) The misfit pooled over the flat tracks on each CCD, minimized over t_c and $\sigma_{\max}$ at each σ_d and shown relative to its minimum, which lies at $\sigma_d = 1.0 \mu\text{m}$ on every CCD. (b) The pooled best-fit t_c and $\sigma_{\max}$ at each σ_d : adding the extra spread lowers t_c and raises $\sigma_{\max}$, since some of what the two-parameter fit attributed to the field-free layer was this floor. (c) The σ_d preferred by each track on its own, which is spread over 0 to $1 \mu\text{m}$: the effect is modest for any one track and consistent across them.

from the depletion thickness of the model, which is the quantity the companion noise model needs. Deeper in the sensor the tracks see about a micron of spreading that the model neglects. It is about what the drift-time diffusion across the depleted thickness plus the micron-scale ionization column of the track itself should give, and it is small compared with the back-surface spread, so it has little effect on UV imaging, but it does bias a two-parameter fit: the fitted t_c of 0.45 on SJI is larger than the model's 0.38 partly because the wedge is stretched to absorb the floor.

The FUV1 CCD is narrower than the other two, at the back surface and in the fitted $\sigma_{\max}$ alike. We have no explanation beyond chip-to-chip variation of the field-free thickness, which depends

on the resistivity of the wafer and the applied bias; the same measurement on the Atmospheric Imaging Assembly (AIA) CCDs, which are of the same design, would show whether such variation is common.

Two systematics deserve comment. First, the tracks are left by protons that deposit thousands of electrons per row, whereas a UV photon liberates one to three. In thick, fully depleted sensors, slow protons leave tracks visibly widened by the mutual repulsion of that charge,¹¹ and if the same occurred in the field-free layer here the tracks would overestimate the single-photon kernel. The tracks themselves can test this, by comparing the kernels of faint and bright tracks, and that comparison is the first thing to add. Second, the depth assignment assumes every flat track crosses the full thickness of the sensor; a track that entered or left through the edge of the field-free layer, or that was clipped by the Bragg cut at one end, would have its depths compressed. The agreement between the model-free and parametric profiles, and between campaigns, suggests such tracks are rare, but a validation on synthetic tracks injected into real frames is the way to bound it.

5 Conclusion

Glancing particle tracks measure the depth-dependent charge-diffusion kernel of a back-illuminated CCD in orbit, with no laboratory access, and on the IRIS CCDs they confirm the field-free model at the back surface while revealing a small residual spread inside the depletion region.

Appendix A: The tracks as the finder cut them out

The cutouts distributed with this article are seven pixels wide and follow the fitted centerline, so they show the tracks after the finder has done its work. Figure 8 shows the longest of the flat tracks and one track that the Bragg cut rejects. Sub-pixel structure is invisible at this scale, which is why the wedge signature has to be extracted by fitting.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper. The analysis code and the text of this article were drafted with the assistance of Claude, a large language model, under the direction and review of the authors.

Code, Data, and Materials Availability

This article is executable:¹² every figure, table, and number quoted in the text is computed from the tracks when the article is built. The code, the track cutouts, and the fits are available at <https://github.com/roytsmart/ccd-diffusion-paper>, and the level-1 IRIS images they were extracted from are available from the IRIS data archive.

Acknowledgments

This work was supported by NASA through the IRIS mission.

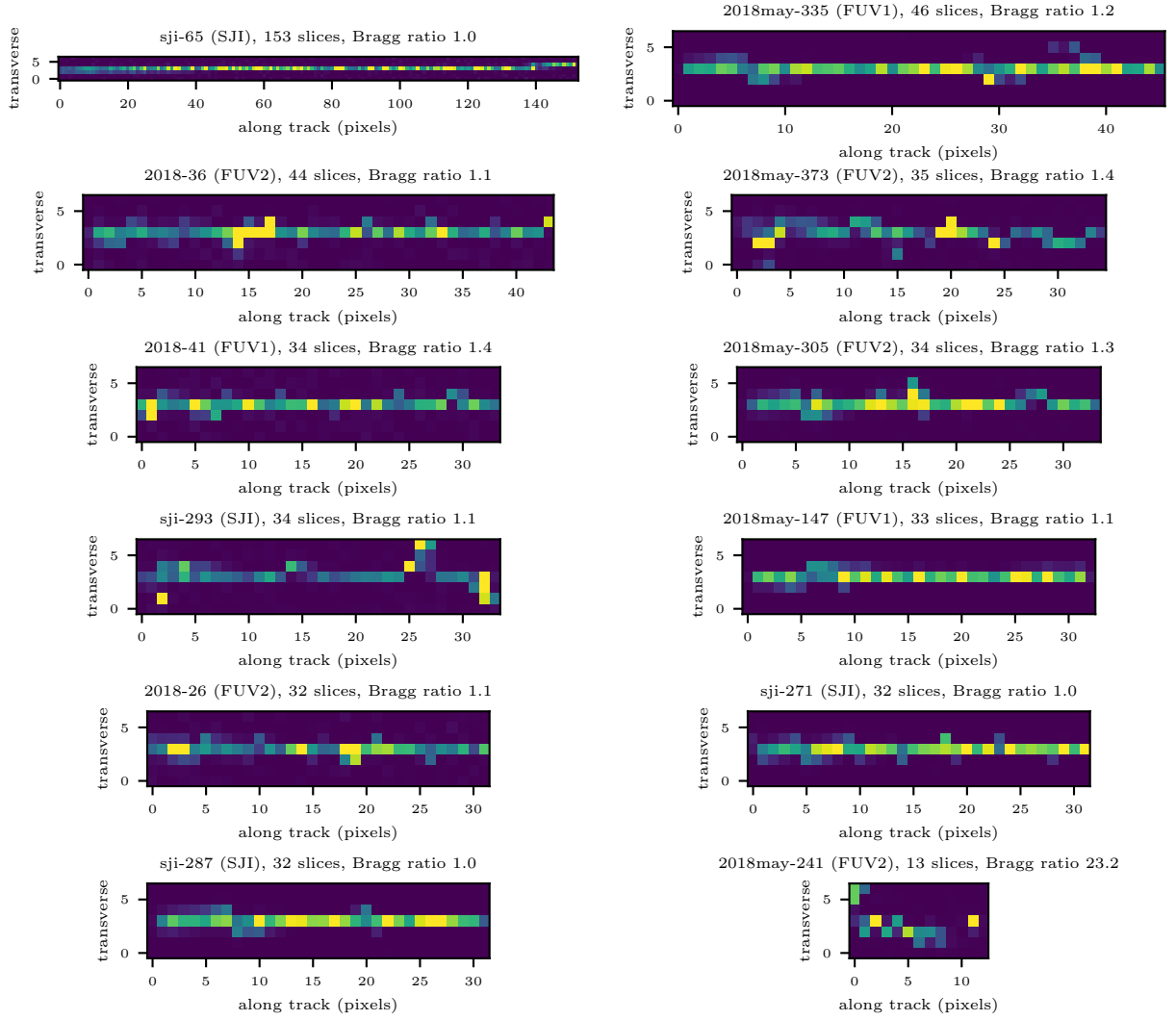


Fig 8 The eleven longest flat tracks and, in the last panel, the track with the strongest Bragg rise, as the finder cut them out: seven pixels wide, re-centered on the integer part of the fitted centerline in every slice, and oriented with the back surface at the left. Each panel's color scale is clipped at its 98th percentile so that a single bright pixel does not hide the track. The stopping track's charge per slice climbs by a factor of twenty toward one end, and the selection of Section 2 rejects it.

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